

Elite Material Co.

The dielectric bottleneck of the AI server: high-speed copper-clad laminate, priced as

2383.TW : Taiwan Stock Exchange | Electronic Materials / PCB Supply Chain

1 EXECUTIVE SUMMARY

Market cap: NT\$1.70-1.90tn (approx. US\$52-59bn) -- range reflects source and date dispersion; stock is highly volatile

Gross / operating margin: FY2025 29.8% / 20.3%; Q2 2026 gross margin 33.9% (+4.5pp QoQ)

Share count: 358.3m shares outstanding (approx. 359m diluted); +1.5% YoY -- no material dilution overhang

Revenue: FY2025 NT\$94.3bn (approx. US\$2.9bn), +46.4% YoY; Q2 2026 NT\$47.3bn, +110% YoY, +43% QoQ

Guidance: Full production, full sales posture reaffirmed for the coming year; monthly capacity approx. 7.5m sheets by end-2026 (approx. +70% vs. 2024), targeted to double again in 2027; price increases of 10%-plus implemented from Q2 2026; M9 sampling 2H 2026, volume 2027

HQ: Guanyin District, Taoyuan City, Taiwan

Elite Material Co. (EMC, Taiwanese ticker 2383.TW, Chinese name Tai Guang Dian) sits one layer further upstream than almost anything else in the artificial intelligence (AI) hardware complex. It does not make chips, servers, or circuit boards. It makes the sheet of material that circuit boards are cut from -- copper-clad laminate (CCL), a sandwich of copper foil bonded under heat and pressure to a resin-and-glass-fibre insulating core. Every graphics processing unit (GPU) baseboard, every 800G and 1.6T network switch, every accelerator tray in a modern AI rack is etched out of this material. EMC's customers are the printed circuit board (PCB) fabricators -- Ibiden, Gold Circuit, Wus, Victory Giant, Unimicron and peers -- who in turn ship to the original design manufacturers (ODMs) building for NVIDIA, Google, Amazon Web Services (AWS) and Meta.

The core thesis is a mix-and-price story, not a volume story. EMC entered the high-speed CCL segment only in 2017 and reached global number one in that niche by 2023; in July 2026, market research firm Prismark ranked it the world's largest CCL maker by revenue for full-year 2025, displacing China's Shengyi Technology and Hong Kong's Kingboard Holdings, who between them had held the top two slots for a decade on commodity volume. The ascent happened because the AI server upgrade cycle is repricing the material itself: as signalling rates climb from 56 to 112 to 224 gigabits per second (Gbps) per lane, the laminate must move up a grade ladder (M6 to M7 to M8 to M9) where each step multiplies the material content value of the same physical board. The investable question is not whether demand exists -- EMC is sold out -- but whether the pricing power and the 12-to-18-month qualification moat survive the 2027 capacity wave that EMC and every competitor are simultaneously building.

2 TECHNOLOGY DEEP-DIVE

ELI5 — What copper-clad laminate actually is

Imagine a very thin, very flat sheet of fibreglass soaked in hardened glue, with a layer of copper foil pressed onto both faces like the foil on a chocolate bar. That is copper-clad laminate. A circuit board factory buys these sheets, dissolves away the copper it does not want with acid, and what is left behind are the copper wires. Then it stacks 30 or 40 of these sheets on top of each other and squeezes them together into one solid board. Now imagine sending a signal down one of those copper wires. Some of the energy arrives at the other end; some of it leaks into the fibreglass-and-glue layer and turns into heat. At slow speeds this leakage does not matter. At AI-server speeds, the leakage eats the signal before it arrives. EMC's business is making the glue and the glass leak less.

At the physics level, the property that matters is the dissipation factor (Df), also called the loss tangent. Df measures the fraction of electromagnetic energy that a dielectric converts into heat rather than propagating forward, and it scales with frequency: insertion loss in decibels per inch is roughly proportional to the product of frequency, the square root of the dielectric constant (Dk), and Df. Standard FR-4 laminate -- the

flame-retardant epoxy-glass workhorse the electronics industry has used for forty years -- has a Df of approximately 0.020 at 10 gigahertz (GHz). That was tolerable when serialiser/deserialiser (SerDes) links ran at 1 to 5 Gbps. It is fatal at 56 Gbps and above, where a channel built on FR-4 would burn its entire loss budget within a few inches of trace and the receiver would see noise rather than data.

Reducing Df requires attacking three materials simultaneously, and this is why the problem is hard rather than merely expensive. First, the resin: conventional epoxy contains polar chemical groups (hydroxyl, ester) whose molecular dipoles rotate under an alternating field and dissipate energy as heat. Low-loss systems substitute non-polar hydrocarbon backbones -- polyphenylene ether (PPE, also written PPO), styrene-butadiene and cyanate-ester chemistries -- which have fewer rotatable dipoles but are correspondingly harder to process, adhere poorly to copper, and cure with tighter thermal windows. Second, the glass: standard E-glass (electrical-grade borosilicate) has a Dk near 6.6, which both raises loss and creates the fibre-weave skew problem -- a trace running over a glass bundle sees a different propagation velocity from a trace running over a resin-rich gap, and at 224 Gbps that intra-pair skew closes the eye diagram. Low-Dk glass (Dk approximately 4.6-5.3), low-coefficient-of-thermal-expansion T-glass, and ultimately Q-glass (fused quartz, Dk approximately 3.7) are the escape routes. Third, the copper: at these frequencies current crowds into roughly the outermost micron of the conductor (the skin effect), so any surface roughness lengthens the effective path and adds loss. M8 and M9 grades require HVLP (high very-low-profile) copper foil with surface roughness below approximately 0.5 microns -- but smooth foil also adheres less well to resin, so peel strength must be recovered chemically rather than mechanically.

EMC formulates its own resin systems in-house. That is the single most important structural fact about the company: the glass and the copper foil are bought, the chemistry is proprietary. Manufacturing then runs through impregnation (glass cloth drawn through a resin bath and B-staged in a treater tower to a tacky semi-cured prepreg), lay-up, and hydraulic vacuum press lamination at temperatures above 200 degrees Celsius for multi-hour cycles during which the resin flows, wets the fibre, and fully cross-links to the rigid C-stage. Yield is governed by voids, delamination, resin-content uniformity and thickness control across a sheet roughly one metre on a side -- process know-how that does not transfer with a datasheet.

Grade tier	Typical Df at 10GHz	Materials set	Where it goes	Competes against
Mid-loss / low-loss (M4-M6 class), including halogen-free FR-4 upgrades	approx. 0.008-0.014	Modified epoxy, E-glass, standard or VLP foil	Handsets, automotive, general-purpose servers, legacy networking	Kingboard, Shengyi, Nan Ya, ITEQ -- highly price-competitive
Ultra-low-loss (M7 class)	below approx. 0.005	PPE / hydrocarbon resin, low-Dk glass, VLP-to-HVLP foil	Mainstream AI server boards, 800G switch line cards	Panasonic Megtron 7, TUC, ITEQ, Doosan, Shengyi high-end lines
Extreme-low-loss (M8 / M8+ class)	approx. 0.0012-0.0015 (roughly 25% below M7)	Advanced PPE blends, low-Dk gen-2 glass, HVLP3/HVLP4 foil	NVIDIA GB200 / GB300 baseboards, densest 1.6T switching, hyperscaler custom ASIC boards	Panasonic, TUC; Chinese peers largely still qualifying
Next-generation (M9 / M9+ class)	approaching 0.001	Requires Q-glass (quartz) rather than E-glass; HVLP4-plus foil	224 Gbps-per-lane platforms; NVIDIA has specified M9 for the 78-layer orthogonal backplane of Rubin Ultra	Panasonic, Resonac, Mitsubishi Gas Chemical, Doosan, TUC -- field narrows sharply
HDI / mSAP materials	n/a (process-defined)	Fine-line-capable resin systems for modified semi-additive process	Smartphone mainboards (adopted by Apple from 2017); increasingly AI modules using HDI build-ups	Panasonic, Shengyi; EMC claims number one for 20 consecutive years, share above 70%
IC substrate-grade CCL	n/a	Substrate-class dimensional stability and thin-core capability	Package substrates -- currently below 1% of EMC revenue, in market-development phase	Ajinomoto build-up film ecosystem, Mitsubishi Gas Chemical, Shengyi, Nan Ya

Grade tier	Typical Df at 10GHz	Materials set	Where it goes	Competes against
Prepreg (bonding sheets)	matched to core grade	B-stage semi-cured resin/glass, sold alongside laminate	Sold as a matched pair with every CCL grade; boosts wallet share per board	Every CCL maker; matched-set supply is a mild lock-in mechanism

Note on nomenclature: the M-series grade ladder is an informal industry shorthand derived from Panasonic's MEGTRON product family and now used generically across the supply chain. EMC markets its own EM-prefixed product codes against each tier; the mapping of specific EMC part numbers to grades is not consistently disclosed in public sources and is therefore presented here at tier level rather than by part number.

ELI5 — Why this cycle changes the demand profile, not just the volume

Think of a restaurant that used to sell only burgers. Demand for meals goes up 30%, which is nice. But at the same time customers stop ordering burgers and start ordering steak. The kitchen cooks roughly the same number of plates, but revenue per plate is several times higher and the margin per plate is higher too, because only three kitchens in the world can cook that steak. That is what is happening to EMC. AI boards are bigger and thicker (more plates), but the far bigger effect is that the material on each board is being upgraded to grades that cost many times more per square metre.

Quantifying the shift. Three independent intensity vectors compound. First, volume per box: an AI server consumes roughly five to seven times more CCL by area than a conventional two-socket server, because of larger baseboards, separate switch trays, and mid-plane assemblies. Second, layer count: the average AI server PCB has gone from approximately 18 layers in 2023 to approximately 32 layers in 2025, a 78% increase in two years, and NVIDIA's Rubin Ultra backplane runs to 78 layers. Every additional layer is another sheet of laminate plus another sheet of prepreg. Third, and dominant, grade mix: brokers covering the name estimate that the CCL content value of a single high-end AI board rises from roughly US\$55 at commodity grade to roughly US\$450 at M9 -- an order-of-magnitude uplift achieved with zero change in unit shipments. Treat the US\$55-to-US\$450 figures as directional sell-side estimates rather than disclosed pricing, but the direction is corroborated by observable price action: high-end CCL grades rose 20% to 40% in the first half of 2026, EMC implemented across-the-board increases of at least 10% from Q2 2026, and industry lead times for specialty laminate extended from a historical 8-12 weeks to 20-30 weeks.

The financial signature of this is visible in the accounts. FY2025 revenue grew 46.4% while gross margin expanded 192 basis points to 29.8%; in Q2 2026 revenue grew 110% year-on-year while gross margin jumped 450 basis points sequentially to 33.9% and net margin reached approximately 20.9%. A pure volume boom does not produce that margin slope. Mix migration plus a seller's market does. Management's own disclosure frames the mix: infrastructure applications are approaching 80% of revenue in 2026, versus server-related revenue of roughly 44% of the total (of which AI servers were roughly 24%) as recently as Q3 2025, with M7-and-above grades already above 60% of revenue at that point.

Position within specific platforms matters for modelling. On NVIDIA's universal baseboard (UBB) EMC is understood to be a second source with roughly 20-25% allocation; on Google's next-generation tensor processing unit (TPU) programme, broker estimates have put EMC's M8 CCL share at 55-60%. These are channel-derived estimates and should be treated as such, but they explain why EMC's revenue is levered to the custom-ASIC (application-specific integrated circuit) cohort as much as to NVIDIA merchant GPUs -- a diversification that is under-appreciated relative to the market's habit of treating the whole PCB chain as an NVIDIA derivative.

3 SUPPLY CHAIN MAPPING

ELI5 — EMC's position in the chain

EMC is the sandwich shop, not the farm. It buys three things: bread (glass fibre cloth), butter (resin chemicals) and ham (copper foil). It makes the sandwich and sells it to the circuit board factory. EMC's own recipe -- the butter -- is the secret it controls. The bread and the ham it must buy, and right now the two or three farms that grow the good bread cannot grow it fast enough. That is simultaneously EMC's biggest cost risk and its biggest competitive protection: if the good bread is scarce, the shops that already have long-standing supply contracts win, and new shops cannot open.

Input	Principal suppliers	Single-source?	Risk note
Low-Dk and low-CTE glass cloth (T-glass, NE-glass, gen-2 low-Dk)	Nittobo (Nitro Boseki, Japan), PPG (US), Taiwan Glass (1802.TW), Nan Ya (1303.TW), Baotek (5340.TWO, a Nittobo subsidiary), Fu-Chiao (1815.TW)	Effectively duopolistic at the top end	Nittobo and PPG together hold above 80% of high-end electronic glass cloth; Nittobo alone is reported above 90% of T-glass. Nittobo expansion does not come online until 2027 and adds only approx. 20%. Prices rose up to 20% in a single step during 2025; industry commentary points to a further 20-30% in 2026 and tightness persisting into 2H 2027. This is the single hardest constraint on EMC's output.
Q-glass (quartz fibre) -- required for M9	Shin-Etsu Quartz, AGC, Te Hung (quartz specialist, Taiwan), a small Chinese base	Yes, in practice	Quoted at roughly US\$80-100 per kilogram versus a small fraction of that for E-glass. A potential 2027 supply gap of above 60% has been flagged by Taiwanese brokers. Q-glass scarcity has reportedly already forced some boards in NVIDIA's Rubin ecosystem to be respecified down to M8 or M7 equivalents. This is the binding constraint on the M9 ramp -- for EMC and for every competitor equally.
HVLP copper foil (HVLP3 / HVLP4, roughness below approx. 0.5 micron)	Mitsui Mining and Smelting, Furukawa Electric, Co-Tech Development (8358.TW), Nan Ya, Lotte Energy Materials	Multi-sourced in name, concentrated in practice	Taiwanese broker work estimates HVLP4 supply gaps of approximately 48% in 2026 and 43% in 2027 on unstable yields. Copper foil is also the input most exposed to the LME copper price, layering a commodity beta on top of a scarcity premium.
Specialty resins (PPE/PPO, hydrocarbon, cyanate ester, BT)	SABIC, Mitsubishi Gas Chemical, Asahi Kasei, Chang Chun Group (Taiwan) for epoxy backbone	No -- and formulation is internal	EMC compounds its own resin systems, so the exposure is to monomer and oligomer feedstock rather than to a finished competitor product. This is the least concentrated of the four key inputs and the main reason EMC captures more of the value chain than a pure converter would.
Silica and inorganic fillers (low-alpha spherical silica)	Denka, Admatechs, Sibelco, Novoray (688300.SS), Lingwe (301373.SZ)	No	Low-alpha-emission grades matter more for substrate-class products than for core CCL; secondary cost item but a qualification gate for the substrate ambition.
Lamination presses and treater lines	Specialist Japanese, Taiwanese and Chinese capital equipment makers	Concentrated	Reported lead times of up to two years. EMC has already deferred approximately 1.2m sheets per month of Kunshan and Zhongshan capacity from Q4 2026 to Q1 2027 on equipment delivery slippage. This cuts both ways: it delays EMC's own ramp but equally locks out would-be entrants.

Geographic concentration. The upstream is overwhelmingly Japanese at the technology frontier (Nittobo, Mitsui, Furukawa, Shin-Etsu, Mitsubishi Gas Chemical, Resonac) with a Taiwanese second tier (Taiwan Glass, Co-Tech, Nan Ya, Fu-Chiao, Baotek) that is being deliberately cultivated by customers as a de-risking measure. EMC's own manufacturing sits in Taiwan (Guanyin, Dayuan), China (Kunshan, Zhongshan, Huangshi), Malaysia (Penang) and the United States. Management asserts EMC is the only Asian CCL supplier with both production and sales presence in the US, and the board approved a further US\$66.8m (approximately NT\$2.2bn) investment in US subsidiary facilities on 29 July 2026 -- modest in absolute terms but strategically significant given the direction of US content rules for domestic AI infrastructure.

Input cost sensitivity is the live margin question. Management flagged at the March 2026 investor day that copper foil and glass fabric prices had risen by more than 30% and that a price increase of at least 10% would be pushed through from Q2 2026. The Q2 print settles the first round of that argument in EMC's favour: gross margin expanded 450 basis points sequentially, meaning realised price plus mix outran input inflation by a wide margin. The forward risk is that the pass-through is a one-shot repricing while glass and quartz inflation is a multi-year ramp, and that the next increase meets more customer resistance now that PCB makers themselves are being squeezed. A second, quieter margin drag is internal: 2026 depreciation

steps up to approximately NT\$2.8bn, roughly NT\$1.0bn above 2025, diluting operating margin independent of gross margin.

4 COMPETITIVE LANDSCAPE

Company	HQ	Revenue	Share	Key differentiator	Weakness vs. EMC
Elite Material (2383.TW)	Taoyuan, Taiwan	FY25 NT\$94.3bn (approx. US\$2.9bn); 1H26 NT\$80.3bn	Approx. 18.9% of all CCL (TPCA/ITRI); 22.1% of specialty laminate; 38.6% of high-speed CCL; above 70% of HDI	In-house resin chemistry plus number-one position in both high-speed and HDI, allowing it to serve AI boards that combine both; only Asian CCL maker with US production	-- (subject company)
Shengyi Technology (600183.SS)	Dongguan, China	CCL sales approx. US\$2.47bn in 2025 (Prismark); group revenue higher	No. 2 globally by CCL revenue; approx. 14% in 2023 rankings	Deep vertical integration and the default beneficiary of Chinese domestic substitution policy; Q1 2026 net profit more than doubled year-on-year	Mix still skewed to mainstream grades; constrained access to Japanese Q-glass and HVL4 for the very top tier; narrower qualification footprint at US hyperscalers and NVIDIA
Kingboard Laminates / Kingboard Holdings (1888.HK / 148.HK)	Hong Kong / China	CCL sales approx. US\$2.31bn in 2025 (Prismark)	No. 3 globally; led rankings at approx. 14.6% in 2023	The most vertically integrated player in the industry -- owns copper foil, glass fabric, resin and downstream PCB capacity, giving it the lowest cost base in commodity grades	Volume-and-cost model with limited presence at M7 and above; the repricing of the market toward specialty grades works directly against its positioning
Nan Ya Plastics (1303.TW)	Taipei, Taiwan	CCL is a modest slice of a large petrochemical group	No. 4 globally in rigid laminate	Owns glass fabric, copper foil and resin internally; strategic weaving partnership with Nittobo secures non-merchant NE yarn allocation	CCL is not the group's centre of gravity; high-speed grade portfolio lags EMC by roughly a generation; slower to reprice
Taiwan Union Technology / TUC (6274.TWO)	Taoyuan, Taiwan	FY25 NT\$30.3bn, +31.5%; 5M26 NT\$19.5bn, +80% YoY; FY25 EPS NT\$12.13	Approx. 10.3% in 2023 Prismark rankings	The credible Taiwanese number two in high-speed; strong at NVIDIA and in switching; accelerating faster than EMC on a percentage basis off a smaller base in 2026	Roughly one-third of EMC's revenue; no comparable HDI franchise; narrower product breadth and thinner balance sheet to fund a 2027 capacity race
ITEQ (6213.TW)	Taoyuan, Taiwan	FY25 NT\$33.1bn; FY25 EPS NT\$4.16; 1H26 NT\$20.7bn, +25.6%	Top-ten global; single-digit share	Broad halogen-free portfolio, established Thai capacity build-out for China-plus-one sourcing	The clearest evidence of divergence: ITEQ's Q1 2026 EPS actually fell year-on-year while EMC's doubled -- its mix is not levered to the AI grades
Panasonic (6752.T) -- MEGTRON	Osaka, Japan	Not separately disclosed	Small revenue share, outsized specification share	The reference standard: the M-grade nomenclature the whole industry uses is derived from MEGTRON. Still the benchmark for qualification	Capacity-constrained and premium-priced; has ceded volume to Taiwanese suppliers who can scale; conglomerate structure limits reinvestment appetite
Doosan Electro-Materials (000150.KS)	Seoul, South Korea	Not separately disclosed	Rising into the top tier	Fast-improving high-speed portfolio with visible NVIDIA-ecosystem engagement; Korean upstream foil support	Sub-scale relative to EMC in both revenue and installed capacity; later to the M8 qualification cycle

The moat has four layers, in ascending order of durability. First, proprietary resin formulation -- real but the most copyable over a five-year horizon. Second, process yield at 40-plus-layer lamination without voids or delamination, which is tacit know-how accumulated through failure. Third, and most important, the qualification cycle: when a server or switch original equipment manufacturer (OEM) designs a platform, it selects and qualifies CCL early, and that process -- engineering validation, signal integrity characterisation, thermal cycling, multi-team sign-off -- takes 12 to 18 months. Once qualified, the material cannot practically

be swapped mid-generation, because a substitute must restart the whole cycle using engineering resources the OEM does not have spare during a volume ramp. EMC's current design wins are therefore not revenues that end when a cheaper rival appears; they are revenues protected for the life of the platform. Fourth, the co-engineering flywheel that follows from incumbency: because EMC is the leader, silicon and system designers work with its R&D two to three years ahead of production, so the next qualification is won before the current one ships in volume.

The most credible threat is not a Chinese cost attack -- that attacks the segment EMC has already exited. It is Shengyi Technology combined with a Chinese domestic Q-glass and HVLP4 breakthrough, aimed specifically at the next qualification window rather than the current one. Shengyi has the balance sheet, the state-backed customer pull from Huawei, Alibaba and Baidu accelerator programmes, and the explicit national policy tailwind; what it lacks is access to the Japanese quartz and ultra-smooth foil supply that M9 requires. If Chinese suppliers close that upstream gap during 2027-2028 -- and Novoray and Lingwe are already qualifying into Shengyi and Nan Ya New Material on the filler side -- then Shengyi can contest the M10 generation on equal technical terms with a structurally lower cost base. The secondary threat is more prosaic and nearer-term: TUC is growing faster than EMC in percentage terms in 2026 and is the natural dual-source partner every OEM wants for exactly the allocation-risk reasons that make EMC's position lucrative. Customers have a strong incentive to build TUC into a genuine second source, and they are the party paying for it.

5 MANAGEMENT AND GOVERNANCE

Chairman and President: Tung Ting-Yu (Dong Dingyu). Tung has run EMC as chairman since June 2016, having previously served as vice chairman, and the board additionally appointed him president (chief executive) in February 2024 -- consolidating both roles in one person. He is a long-tenured operator rather than a professional manager parachuted in for the AI cycle, and Taiwanese business press profiles him as an unusually hands-on, strategy-focused executive with a reputation for operational severity. The predecessor chairman, Tsai Hui-Liang, moved to vice chairman in the same 2016 transition and remains on the board. Board composition has been progressively professionalised over two decades; the 2001 director slate was populated with politically connected retired officials, and that character has been removed.

Guidance and execution track record is the strongest single argument for the management assessment. Three calls made publicly and then delivered: the March 2026 statement that a price increase of at least 10% would be implemented from Q2 to offset a 30%-plus rise in copper foil and glass costs (Q2 gross margin expanded 450 basis points sequentially -- delivered, and then some); the guidance that M8 would enter the 1.6T switching market with shipments from Q1 2026 (delivered); and the repeated full-production-full-sales characterisation, which has held since 2024. Management was also candid, unprompted, about the negatives -- explicitly disclosing the deferral of 1.2m sheets per month of Kunshan and Zhongshan capacity from Q4 2026 to Q1 2027 on equipment lead times, and quantifying the 2026 depreciation step-up to approximately NT\$2.8bn and its dilutive effect on operating margin. That is not the behaviour of a management team managing a narrative.

Capital allocation has been disciplined and, notably, organic. There is no acquisition history of consequence, no buyback programme, and no large equity issuance -- share count has grown only about 1.5% year-on-year, consistent with employee compensation rather than dilutive financing. Dividends have been paid on a stable policy of roughly 56-61% of earnings for five consecutive years, rising with earnings rather than being smoothed: NT\$8.5, NT\$10, NT\$17 and NT\$25 per share against fiscal 2022 through 2025 respectively. The capital that is not returned is going into capacity: trailing-twelve-month capital expenditure of approximately NT\$12.6bn against operating cash flow of approximately NT\$12.5bn, funded partly with an incremental NT\$9.6bn of net borrowing. This is a deliberate decision to trade near-term free cash flow for share of a capacity-constrained market, and given that the constraint is equipment lead times rather than capital availability, it is the correct trade. It does mean free cash flow is currently approximately zero and

the company carries a small net debt position.

Ownership and governance. Insider ownership is modest and the disclosures conflict by definition: aggregated director-and-officer holdings were reported at approximately 9.2% of shares outstanding in a 2023 filing, while aggregator data using a narrower insider definition shows approximately 1.94%. Institutional ownership is approximately 50%, with foreign institutions a large component. There is no dual-class share structure and no disclosed related-party transaction of significance. The governance items worth flagging are: (i) the combination of chairman and president in a single individual since February 2024, which weakens board oversight of the executive and creates succession concentration -- the succession question is being asked openly in the Taiwanese press; (ii) a small amount of director share pledging disclosed historically (roughly 3% of directors' holdings), immaterial at present but a monitoring item in a stock this volatile; and (iii) low absolute insider economic alignment relative to the market value now at stake. On balance: competent and credible, demonstrably accurate on guidance, capital-allocation-disciplined, but with a single-point-of-failure leadership structure.

6 FINANCIAL TRAJECTORY

Fiscal year (NT\$m unless noted)	FY2021	FY2022	FY2023	FY2024	FY2025
Revenue	38,500	38,673	41,296	64,377	94,261
Revenue growth	+41.5%	+0.5%	+6.8%	+55.9%	+46.4%
Gross profit	10,069	9,710	11,333	17,970	28,120
Gross margin	26.2%	25.1%	27.4%	27.9%	29.8%
Operating income	6,923	6,225	7,346	12,152	19,108
Operating margin	18.0%	16.1%	17.8%	18.9%	20.3%
EBITDA	7,616	6,972	8,618	13,834	20,858
Net income	5,493	5,073	5,488	9,578	14,649
Net margin	14.3%	13.1%	13.3%	14.9%	15.5%
Diluted EPS (NT\$)	16.46	14.86	15.75	27.30	40.81
R&D expense	771	953	1,267	1,747	2,142
Dividend per share (NT\$)	10.00	8.50	10.00	17.00	25.00
Free cash flow	1,551	1,005	-391	1,397	2,478

Quarter (NT\$m unless noted)	Q1 2025	Q2 2025	Q3 2025	Q4 2025	Q1 2026	Q2 2026
Revenue	21,683	22,504 (est.)	25,146	24,928	33,067	47,270
Revenue YoY	+68%	n/d	+26%	+34%	+53%	+110%
Revenue QoQ	+17%	+4% (est.)	+12%	-0.9%	+33%	+43%
Gross margin	30.4%	n/d	30.1%	28.6%	29.4%	33.9%
Net income	3,470	3,477 (est.)	3,965	3,737	5,339	9,870
Net margin	16.0%	15.4% (est.)	15.8%	15.0%	16.1%	20.9%
EPS (NT\$)	10.01	10.03 (est.)	11.19	10.44	14.89	27.55

Q2 2025 figures are derived by subtraction from disclosed nine-month cumulative totals rather than taken from a primary disclosure, and are marked accordingly. Q4 2025 revenue is derived from the disclosed full-year total less the nine-month cumulative.

The shape of the series is what matters. Revenue was flat from 2021 to 2023 -- this was a cyclical materials business with a mid-20s gross margin and a low-to-mid-teens net margin, trading like one. The inflection begins in 2024 and the second derivative is still positive: sequential revenue growth of 33% in Q1 2026 and 43% in Q2 2026, with the first genuine margin break in Q2 2026 as the April price increase landed in full. First-half 2026 cumulative EPS of NT\$42.46 already exceeds full-year 2025 EPS of NT\$41.67. Q2 2026 marked the sixth consecutive quarter in which quarterly EPS exceeded one full share par value -- an idiosyncratically Taiwanese metric, but a useful marker of the persistence of the run.

Guidance and medium-term targets. Management has reaffirmed the full-production-full-sales posture for the coming year and named AI servers, general-purpose servers, switches, low-earth-orbit satellite systems and memory modules as growth vectors. The capacity roadmap is the key quantitative guidance: approximately 7.5m sheets per month by the end of 2026, roughly 70% above 2024 levels, with a stated intention to double again from that 2026 base during 2027. Note that the March 2026 investor day framed the 2027 exit target as 9.45m sheets per month with 3.3m added during 2027 across Taiwan, China and Malaysia -- these two framings are not obviously reconcilable and the discrepancy is flagged in the verification log rather than resolved. Near-term capacity additions comprise 300k sheets per month at Zhongshan from Q1 2026 and 300k at Dayuan in Q4 2026, with 1.2m sheets per month at Kunshan and Zhongshan pushed from Q4 2026 into Q1 2027 on equipment delivery slippage. Substrate-grade CCL remains below 1% of revenue and is explicitly in a market-development phase; the new Dayuan line is designed to switch flexibly between substrate grades and M7-to-M9 CCL, which is a sensible hedge on where the mix lands.

Balance sheet items that matter. Cash of approximately NT\$22.1bn against total debt of approximately NT\$25.2bn gives a small net debt position of roughly NT\$3.1bn -- trivial against annualised EBITDA now running well above NT\$30bn, with interest coverage above 40 times and debt to EBITDA near 1.0 times. Equity book value is approximately NT\$47.9bn. The item that deserves attention is not leverage but free cash flow: trailing capital expenditure of approximately NT\$12.6bn consumed essentially all of the approximately NT\$12.5bn of operating cash flow, leaving trailing free cash flow at approximately zero, funded by roughly NT\$9.6bn of net new borrowing alongside a working capital build. Investors underwriting this name on a free-cash-flow basis will not find support; the case rests on earnings power and on the capacity being built now converting into 2027-2028 revenue. There has been no equity raise, no impairment of consequence beyond a small NT\$335m goodwill write-down and NT\$132m asset write-down recorded in FY2025, and no dilution beyond routine employee compensation. Current ratio of 1.33 and quick ratio of 1.01 are adequate but not comfortable, and are the numbers to watch if the receivables and inventory build continues at the Q2 pace.

7 CATALYSTS AND RISKS

Catalyst	Timing	Impact	Probability
M9 limited sampling begins	2H 2026	Confirms EMC is first-to-qualify on the 224 Gbps generation. The M9 grade carries the largest content-value step in the ladder; being first into sampling effectively pre-books the 2027-2028 design-in slots at NVIDIA Rubin-class and next-generation hyperscaler ASIC platforms.	High
Q3 2026 results confirm gross margin above 34%	Late Oct 2026 (est.)	Q2's 33.9% was the first quarter in which the April price increase was fully reflected. A second consecutive quarter at or above that level converts a one-off repricing into a demonstrated new margin base and re-anchors 2027 consensus.	Medium
Dayuan (Taoyuan) line comes online, 300k sheets/month	Q4 2026	Adds volume into an allocation-constrained market where every incremental sheet is pre-sold, and provides substrate-grade optionality. Slippage here after the Kunshan/Zhongshan deferral would be a second execution miss.	Medium

Catalyst	Timing	Impact	Probability
Second across-the-board price increase	2H 2026 - 1H 2027	Glass, quartz and HVLP4 foil inflation is a multi-year ramp, not a step. A successful second increase would prove pricing power is structural rather than an artefact of a single tight quarter.	Medium
US manufacturing footprint expansion executes	2026-2027	The US\$66.8m board approval of 29 July 2026 positions EMC ahead of any domestic-content requirement for US AI infrastructure. Small in capital terms, potentially decisive in allocation terms if US sourcing rules tighten.	Medium
Kunshan and Zhongshan capacity lands in Q1 2027 as re-guided	Q1 2027	1.2m sheets per month, already deferred once. On-time delivery restores confidence in the 2027 capacity doubling that the current multiple is discounting.	Medium
Substrate-grade CCL wins first meaningful qualification	2027+	Currently below 1% of revenue. A genuine entry into package substrate materials would open an adjacent market where Ajinomoto and Mitsubishi Gas Chemical currently hold near-monopoly positions.	Low
Q-glass supply eases enough for a full M9 ramp	2027	The binding constraint on the entire M9 generation. Easing would unlock volumes; continued scarcity caps everyone's M9 revenue and forces board respecification down to M8.	Low

Risk 1 -- The quartz glass ceiling is not EMC's to control. M9 grade requires Q-glass (fused quartz fibre) rather than E-glass, and Q-glass supply is concentrated in a handful of Japanese and specialist producers at roughly US\$80-100 per kilogram. Taiwanese broker work has flagged a potential 2027 supply gap above 60%, and there are already reports that boards inside NVIDIA's Rubin ecosystem have been respecified down to M8 or M7 equivalents because Q-glass could not be secured. If that persists, EMC's M9 revenue simply does not arrive on the timeline the market is discounting -- not because EMC failed, but because Nittobo, Shin-Etsu Quartz and AGC did not expand fast enough. This is the single risk most likely to break the 2027 earnings bridge, and EMC has no lever on it.

Risk 2 -- Shengyi Technology contests the next qualification window, not this one. The qualification moat protects platforms already designed in; it offers nothing on platforms not yet designed. Shengyi has doubled net profit year-on-year, is the designated national champion of Chinese domestic substitution, and has the customer pull of Huawei Ascend, Alibaba and Baidu accelerator programmes to fund a technical assault. The gating item is upstream: Chinese suppliers do not yet have equivalent Q-glass and HVLP4 copper foil. But Novoray and Lingwe are already qualifying spherical silica into Shengyi and Nan Ya New Material, and Chinese glass-fibre and foil capability is improving on a policy-funded timeline. If that upstream gap closes in 2027-2028, Shengyi contests the M10 generation with equal materials access and a structurally lower cost base, and EMC's 38.6% high-speed share becomes the high-water mark.

Risk 3 -- Customers are actively funding a second source, and TUC is it. Taiwan Union Technology grew 5M 2026 revenue 80% year-on-year against EMC's 82% first-half growth, off a base roughly one-third the size, and is the obvious dual-source candidate for NVIDIA and switch OEMs who have just spent two years discovering what single-source allocation risk feels like. The customer has both the motive and the engineering budget to qualify TUC into the next platform generation; EMC's own margin expansion is precisely what creates that motive. Watch TUC's allocation commentary at each print, not EMC's.

Risk 4 -- The price increase was a one-shot and input inflation is a ramp. Q2's 450 basis point gross margin expansion reflects a single April repricing of at least 10% against inputs that had risen more than 30%. Low-Dk gen-2 glass cloth and T-glass are guided by industry sources to rise a further 20-30% during 2026, with tightness persisting into 2H 2027, and HVLP4 foil carries an estimated 43% supply gap in 2027. If EMC's next increase meets resistance -- and PCB fabricators are themselves being squeezed between EMC and their ODM customers -- gross margin compresses back toward 30% even with volumes intact. Layer on the 2026 depreciation step-up to approximately NT\$2.8bn (roughly NT\$1.0bn above 2025) and operating margin can fall while gross margin merely stalls.

Risk 5 -- Equipment lead times cut both ways, and EMC has already slipped once. Lamination press lead times of up to two years forced the deferral of 1.2m sheets per month at Kunshan and Zhongshan from Q4 2026 to Q1 2027. In a sold-out market, deferred capacity is deferred revenue with no offsetting benefit, and a second slippage would call into question the 2027 capacity doubling on which the current multiple depends. The counter-argument -- that the same lead times lock out competitors -- is real but asymmetric: EMC's valuation is priced on its own ramp executing, not on rivals failing.

Risk 6 -- Extreme valuation and positioning leave no room for a soft quarter. The shares rose more than 450% in the twelve months to July 2026 and carry a beta near 1.5 with recent twenty-day realised volatility around 8%. The stock fell nearly 12% in a single week during July 2026 and roughly 6.7% in a single session. At a trailing price-to-earnings multiple in the high seventies and a price-to-book near 40 times, the equity is discounting flawless execution of the 2027 capacity doubling and the M9 ramp simultaneously. A hyperscaler capital expenditure digestion quarter -- the kind that has periodically hit the AI complex -- would de-rate this name far more violently than it would a diversified materials peer, regardless of EMC's own order book.

Risk 7 -- Governance concentration in a single individual. Tung Ting-Yu has been chairman since 2016 and additionally president since February 2024, with insider economic ownership reported anywhere between roughly 1.9% and 9.2% depending on definition. The combination of board chair and chief executive in one person removes a layer of oversight at exactly the moment the company is making the largest capital allocation decisions in its history, and Taiwanese press coverage has begun asking the succession question openly. This is not an acute risk today -- the execution record is strong -- but it is a real concentration in a company now worth north of US\$50bn.

8 VALUATION CONTEXT

Company	Market cap	FY2025 revenue	P/E	Revenue growth	End-market exposure
Elite Material (2383.TW)	NT\$1.70-1.90tn (approx. US\$52-59bn)	NT\$94.3bn (approx. US\$2.9bn)	Trailing approx. 78x (est., on approx. NT\$64 TTM EPS); approx. 49x on 2026 consensus	FY25 +46.4%; 1H26 +82%; Q2 26 +110% YoY	Infrastructure approaching 80% of revenue in 2026 -- AI servers, 800G/1.6T switching, plus HDI in handsets, automotive, LEO satellite
Taiwan Union Technology (6274.TWO)	Not independently verified	NT\$30.3bn	Trailing approx. 120x on FY25 EPS of NT\$12.13; approx. 34-37x on 2026 consensus	FY25 +31.5%; 5M26 +80% YoY; Q1 26 EPS +79% YoY	High-speed CCL for AI servers and switching; narrower product breadth, no HDI franchise
ITEQ (6213.TW)	Approx. NT\$103bn (as at May 2026)	NT\$33.1bn	Approx. 32x trailing (Dec 2025 reference); approx. 34-37x on 2026 consensus	FY25 EPS NT\$4.16, +84%; 1H26 revenue +25.6%; Q1 26 EPS -6% YoY	Broad halogen-free CCL across consumer, networking, automotive; materially lower AI grade mix -- the negative control in this comp set
Shengyi Technology (600183.SS)	Not independently verified	CCL sales approx. US\$2.47bn (Prismark); group revenue higher	Not independently verified	Q1 26 net profit +105% YoY	Broad CCL across Chinese domestic electronics; leveraged to Chinese domestic AI accelerator substitution rather than to US hyperscalers

Trailing EPS for EMC is computed from disclosed quarterly figures rather than sourced from a provider, because published aggregator statistics predate the 29 July 2026 Q2 release. Consensus 2026 multiples are drawn from third-party commentary and dispersion is wide: one domestic brokerage cited 2026 EPS of NT\$77.34 and 2027 EPS of NT\$136.04, but 1H 2026 EPS of NT\$42.46 is already actual, which makes the NT\$77 figure look stale post-Q2. Treat all forward multiples as indicative.

Bull case. The AI grade ladder is a repricing of a physical material, not a volume cycle, and EMC has the only combination in the industry of in-house resin chemistry, number-one share in both high-speed CCL (38.6%) and HDI (above 70%), and a US manufacturing footprint. Q2 2026 proved the pricing thesis:

revenue doubled while gross margin expanded 450 basis points, which is the financial signature of mix migration into a supply-constrained tier, not of a demand spike. Capacity doubles from the 2026 base during 2027 into a market that is sold out with 20-30 week lead times, and M9 -- the largest content-value step yet, with an estimated per-board material value approaching US\$450 against roughly US\$55 at commodity grade -- samples in 2H 2026 for volume in 2027. If 2027 EPS lands near the NT\$136 a domestic broker projects, the shares at roughly NT\$5,000 sit on 37 times 2027 earnings for a business compounding earnings above 80% with a 40% return on equity, a stable 60% payout policy, and a qualification moat that protects each design win for the life of the platform. On those numbers the current multiple is not demanding; it is the fair price of a two-year visibility window.

Bear case. Everything above is already in the price, and then some. The shares are up more than 450% in twelve months on a beta of 1.5, trading near 40 times book and in the high seventies on trailing earnings, with trailing free cash flow at approximately zero because all operating cash flow is being consumed by capital expenditure funded partly with new debt. The margin expansion that underwrites the story rests on a single April repricing against inputs that are guided to inflate a further 20-30% in glass and to run a 43% supply gap in HVLP4 foil during 2027, while 2026 depreciation steps up roughly NT\$1.0bn. Capacity has already slipped once on two-year equipment lead times. The M9 ramp on which the 2027 bridge depends is gated by a Q-glass supply that EMC does not control and that has already forced boards to be respecified downward. Meanwhile customers, having been squeezed, are explicitly funding TUC as a second source, and Shengyi is coming for the generation after next with state backing and a lower cost base. The de-rating triggers are identifiable and near: a single quarter of gross margin below 32%, a second capacity deferral, an M9 sampling delay, or any hyperscaler capital expenditure digestion signal. At this multiple, any one of them is worth 30-40%.

9

WHAT MOVES IT NEXT -- EARNINGS SIGNAL MAP

ELI5 — Why this section exists

You cannot read every filing from every company you follow. This table is a triage tool: for each name, when the next report lands, and the single number in it that will actually move the stock. Everything else in the release is noise for portfolio purposes. Read the one metric, decide, move on.

Company	Next earnings	Days away	The one metric	Why it matters
Elite Material (2383.TW)	Late Oct 2026, Q3 (est.)	approx. 90	Gross margin holding at or above 33.9%, with M9 sampling confirmed as begun	Q2's 33.9% was the first full quarter of the April price increase. A second consecutive quarter at that level converts a one-off repricing into a structural margin base and validates the 2027 bridge; a slip back toward 30% -- especially with the NT\$2.8bn depreciation step-up now running -- reframes Q2 as the peak. Paired with this, any language deferring M9 sampling out of 2H 2026 would directly attack the 2027 content-value step.
Taiwan Union Technology (6274.TWO)	Early-to-mid Aug 2026, Q2 (est.)	approx. 7-14	Whether TUC's gross margin expanded as much as EMC's, and any commentary on new platform allocation wins	This is the cleanest available read on whether EMC's Q2 margin break was industry-wide pricing power or EMC-specific mix. If TUC also expanded 400-plus basis points, the whole tier is repricing and EMC's premium multiple is harder to defend. If TUC lagged, EMC's mix advantage is proven and the premium is earned. Allocation commentary also tests the second-source risk directly.
ITEQ (6213.TW)	Mid-Aug 2026, Q2 (est.)	approx. 14	Whether Q2 EPS returns to year-on-year growth after Q1's decline	ITEQ is the negative control. Its Q1 2026 EPS fell year-on-year while EMC's doubled, which isolates AI grade mix as the driver rather than a general CCL upcycle. If ITEQ's Q2 also lags badly, the thesis that value is concentrating in the top two or three grades is confirmed; if ITEQ inflects sharply, the upcycle is broadening and the specialty premium is at risk of compressing.

Priority order: read TUC's Q2 first. It lands within roughly two weeks, it is free information about EMC, and it discriminates between the two competing explanations for the Q2 margin break -- industry-wide repricing versus EMC-specific mix leadership -- which is the single most important unresolved question in the thesis. ITEQ follows a few days later as the negative control. EMC's own Q3 print in late October is the decision point, but by the time it arrives the Taiwanese peers will have already told you most of what it will say about pricing; what only EMC can tell you is whether M9 sampling started on schedule and whether Dayuan is on track for Q4. Monthly revenue disclosures, which Taiwanese issuers publish around the tenth of each month, provide a useful interim check on whether the Q2 run-rate is holding.

A VERIFICATION LOG

Item	Status	Note
Ticker and exchange (2383, TWSE)	CLEAN	Confirmed across multiple aggregators and Taiwanese sources
FY2021-FY2025 income statement	CLEAN	S&P Global Market Intelligence via aggregator; internally consistent with company disclosures
Q2 2026 results (revenue NT\$47.27bn, EPS NT\$27.55, GM 33.9%, net income NT\$9.87bn)	CLEAN	Released 29 July 2026; cross-checked against two independent reports
Market cap NT\$1.70-1.90tn	FLAGGED	Sources dated 7-10 July 2026 give NT\$1.70tn to NT\$1.90tn; the stock moved more than 12% in a week during July and Q2 results landed 29 July. Presented as a range; verify at trade date
Q2 2025 quarterly figures	EST.	Derived by subtraction from disclosed nine-month cumulative totals, not from a primary quarterly disclosure
TTM EPS approx. NT\$64 and trailing P/E approx. 78x	EST.	Computed from disclosed quarterly EPS; published aggregator statistics predate the Q2 release
Capacity target: 7.5m sheets/month end-2026 doubling in 2027 vs. 9.45m sheets/month end-2027	FLAGGED	Two management framings from different disclosures are not obviously reconcilable. Both reported; neither adopted as the base case
Market share: 18.9% (TPCA/ITRI, all CCL), 22.1% specialty, 38.6% high-speed, above 70% HDI	FLAGGED	Different denominators from different sources. Prismark rigid CCL market of US\$18.7bn in 2025 against EMC revenue of approx. US\$2.9bn implies approx. 16%, not 18.9%. Directional, not precise
Per-board CCL content value US\$55 to US\$450 across the grade ladder	EST.	Sell-side estimate cited in Taiwanese research commentary; not a disclosed price. Direction corroborated by observed 20-40% high-end price increases
NVIDIA UBB share 20-25%; Google TPU M8 share 55-60%	EST.	Channel-derived broker estimates; not disclosed by EMC or by customers
Insider ownership 1.94% vs. 9.2%	FLAGGED	Aggregator narrow-definition figure versus a 2023 Taiwanese filing of aggregated director and officer holdings. Both reported
Q-glass supply gap above 60% in 2027; HVLP4 gap 48%/43% in 2026/2027	EST.	Taiwanese brokerage estimates reported in trade press; directional
Global CCL market US\$21.5bn in 2026, +34.2%	EST.	TPCA/ITRI joint projection; vendor forecast, directional only
Rigid CCL market US\$18.7bn in 2025, +24.4%; SYTECH US\$2.47bn, Kingboard US\$2.31bn	CLEAN	Prismark, June/July 2026 publication
TUC FY2025 revenue NT\$30.3bn	FLAGGED	Two sources give NT\$30.34bn and NT\$30.43bn. Immaterial difference; range used
TUC and Shengyi market caps	UNVERIFIABLE	Not confirmed to a reliable current source within the research budget. Left blank rather than estimated
Next earnings dates for all three names	EST.	Estimated from prior-quarter cadence (EMC reported Q3 2025 on 30 October 2025, Q2 2026 on 29 July 2026). Not confirmed from company investor relations calendars
EMC product part numbers by grade tier	UNVERIFIABLE	EMC markets EM-prefixed codes but public mapping to M-grades is inconsistent. Presented at tier level only

Item	Status	Note
Employee count 572 shown by aggregator	FLAGGED	Almost certainly parent-company-only. Not used anywhere in this report

Corrections applied: the aggregator revenue-per-employee and profits-per-employee statistics were discarded as parent-only. Market cap was converted from a point figure to a range after sources diverged by more than 10%. No other corrections were required.

B

ACRONYM GLOSSARY

Acronym	Full name
ABF	Ajinomoto Build-up Film -- the dominant dielectric film for IC package substrates
AI	Artificial Intelligence
ASIC	Application-Specific Integrated Circuit
AWS	Amazon Web Services
CCL	Copper-Clad Laminate
CTE	Coefficient of Thermal Expansion
Df	Dissipation factor, also called loss tangent -- fraction of signal energy lost as heat
Dk	Dielectric constant, also called relative permittivity
EMC	Elite Material Co. (the subject company)
FR-4	Flame Retardant 4 -- the standard epoxy-glass laminate grade
Gbps	Gigabits per second
GHz	Gigahertz
GPU	Graphics Processing Unit
HDI	High Density Interconnect -- fine-line, micro-via PCB construction
HVLP	High Very-Low-Profile (copper foil, sub-micron surface roughness)
ITRI	Industrial Technology Research Institute (Taiwan)
LEO	Low Earth Orbit
LME	London Metal Exchange
mSAP	Modified Semi-Additive Process -- fine-line PCB patterning method
ODM	Original Design Manufacturer
OEM	Original Equipment Manufacturer
PCB	Printed Circuit Board
PPE / PPO	Polyphenylene Ether / Polyphenylene Oxide -- low-loss resin backbone
Prepreg	Pre-impregnated bonding sheet -- semi-cured (B-stage) resin and glass
SerDes	Serialiser / Deserialiser -- the high-speed interface driving signalling rates
TPCA	Taiwan Printed Circuit Association
TPU	Tensor Processing Unit (Google's custom AI accelerator)
TUC	Taiwan Union Technology Corporation
UBB	Universal Baseboard (the GPU carrier board in AI server platforms)
VLP	Very-Low-Profile (copper foil grade below standard roughness)

C

DISCLAIMER

This document is for internal research purposes only and does not constitute investment advice, a recommendation, or an offer to buy or sell any security. It contains no price target and no buy or sell rating. All data should be independently verified before it is relied upon for any portfolio decision; figures marked EST. or FLAGGED in the verification log are explicitly not confirmed. Financial data is presented in New Taiwan dollars unless otherwise stated, with US dollar conversions at approximately NT\$32.4 to US\$1. The subject company reported quarterly results on 29 July 2026, two days before the date of this report; some third-party aggregator statistics cited herein predate that release. Report date: 31 July 2026.